

THE CENTRAL QUESTION

Does ~135x sales (~\$81B on \$602M FY25 revenue) price a realistic path to a space prime — or a SpaceX-scale outcome a decade of fundamentals can't reach?

Rocket Lab trades at ~\$81B on \$602M FY25 revenue — ~135x sales. Electron (#2 small-lift) + Neutron (reusable medium-lift, Q4 2026 debut) + a vertically-integrated Space Systems segment with a path to owned constellations. The DCF asks what scale, prime margins, and — in the upper scenarios — a Starlink-style constellation second act are plausible over a decade. The finding: launch+systems alone is priced beyond ten years; only if RKLB originates an owned constellation does the ultra-bull clear spot (+159%, a ~\$213B smaller-SpaceX) — even then the modal base leaves the weighted ~63%: the market prices near-certain success.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$52.04 expected fair value today
-63.4% vs spot \$142.00

FORWARD COMPOUNDED VALUE

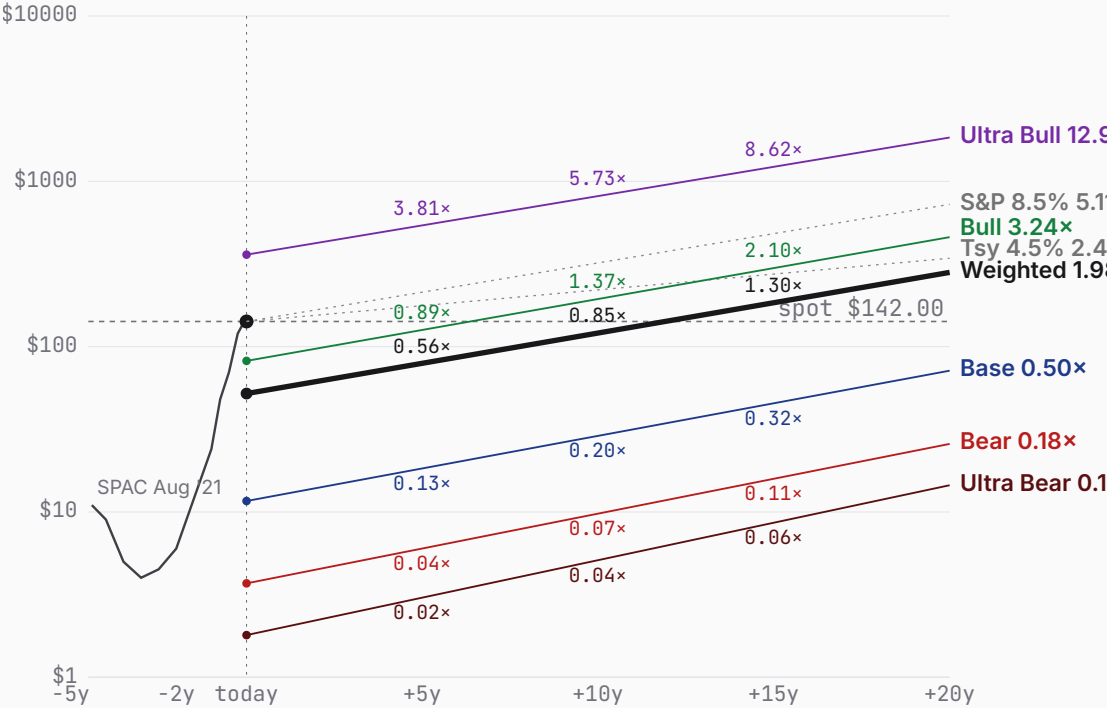
+5y	+10y	+15y	+20y
\$79.31	\$120.92	\$184.42	\$281.36
0.56× spot	0.85× spot	1.30× spot	1.98× spot

WEIGHTING RATIONALE

- Ultra Bear 10%** Neutron fails; dilution; SpaceX runs away.
- Bear 25%** Neutron slips; credible #2 but capped.
- Base 35%** Neutron works; #2 + Space Systems prime ~18%.
- Bull 22%** Reusable Neutron scales; top-3 + owned constellation begins.
- Ultra Bull 8%** Starlink-style constellation; ~\$213B smaller-SpaceX — clears spot.

Bull (22%) + Ultra Bull (8%) together contribute \$46.86 — 90% of the \$52.04 expected value despite 30% combined probability. Today's spot (\$142.00) sits 173% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED



ULTRA BEAR	\$1.80	BEAR	\$3.70	BASE	\$11.65	BULL	\$82.02	ULTRA BULL	\$360.14
	-98.7% vs spot		-97.4% vs spot		-91.8% vs spot		-42.2% vs spot		+153.6% vs spot
TAM 1.1% P(fail) 25% S2C 1.7 Prob 10%		TAM 2.4% P(fail) 12% S2C 1.9 Prob 25%		TAM 4.9% P(fail) 5% S2C 2.1 Prob 35%		TAM 13.8% P(fail) 3% S2C 2.4 Prob 22%		TAM 30.4% P(fail) 2% S2C 2.5 Prob 8%	
Neutron fails; dilution; SpaceX runs away.		Neutron slips; #2 but sub-scale.		Neutron works; #2 prime; 18% margin.		Neutron scales; owned-constellation second act begins.		Starlink-style constellation; smaller-SpaceX by 2035.	

PROBABILITY WEIGHTS

Ultra Bear 10% Bear 25% Base 35% Bull 22% Ultra Bull 8% · Weighted expected \$52.04 (-63.4% vs spot)

ULTRA BEAR	\$1.80	BEAR	\$3.70	BASE	\$11.65	BULL	\$82.02	ULTRA BULL	\$360.14
	-98.7% vs spot		-97.4% vs spot		-91.8% vs spot		-42.2% vs spot		+153.6% vs spot

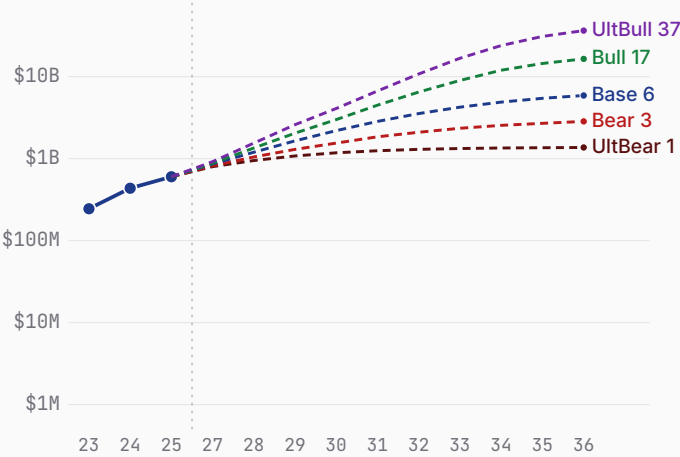
Probability 10%	Probability 25%	Probability 35%	Probability 22%	Probability 8%
Neutron fails; dilution; SpaceX runs away. WHY 10% The compound bear: a Neutron failure (the FY26 tank-test failure is a real datapoint) plus Space Systems deceleration plus persistent thin margins. Aerospace hardware is unforgiving and SpaceX's cost curve is brutal. 25% within-scenario failure probability reflects that RKLБ, while revenue-generating with \$1.85B backlog, is still capital-hungry and Neutron-dependent for the bull thesis. WHAT HAPPENS Neutron's tank-test troubles compound into a failed or indefinitely delayed program; RKLБ stays a sub-scale small-lift + components shop. Launch cadence stalls, Space Systems growth decelerates as satellite primes in-source, and gross margins stay structurally thin against a fixed-cost manufacturing base. Repeated dilutive ATM raises fund the burn (start cash \$0.83B), share count creeps toward 590M, and by FY35 revenue is ~\$1.4B at ~7-8% operating margin. With a 25% probability of an outright restructuring/recap, the expected per-share collapses toward the distress floor — orders of magnitude below today's price.	Neutron slips; #2 but sub-scale. WHY 25% Neutron works but underwhelms on cadence/cost; Space Systems grows steadily without becoming a prime. 12% failure probability — lower than a pre-revenue name given backlog and balance sheet, but Neutron execution risk is real. The 25% weight reflects this as a very plausible 'competent but capped' outcome. WHAT HAPPENS Neutron eventually flies but late and at low cadence; RKLБ remains a credible #2 launch name and a solid Space Systems supplier, but never escapes SpaceX's shadow on cost or volume. Revenue compounds to ~\$2.85B by FY35 at ~12% operating margin — real, but a fraction of the bull trajectory. Modest dilution funds the Neutron ramp. The DCF lands in the low-single-digit dollars per share — the market is pricing a far larger outcome. This is the 'good company, wrong price' world.	Neutron works; #2 prime; 18% margin. WHY 35% Requires the least faith: Neutron works (not dominates), Space Systems keeps compounding, margins reach an aerospace-prime ~18%. No SpaceX-toppling assumption. 5% failure probability given backlog, cash, and revenue diversification. 35% weight as the central outcome. WHAT HAPPENS The modal path: Neutron certifies and reaches meaningful cadence, RKLБ consolidates the clear #2 launch position, and the vertically-integrated Space Systems business scales into a genuine space prime. Revenue compounds ~25% to ~\$5.9B by FY35 at an ~18% mature operating margin — aerospace-prime economics, not software. Light early dilution; the business self-funds by the late 2020s. DCF ~\$12/share. Even this constructive base case sits ~92% below today's \$142 — a measure of how much the market is paying for a larger, faster outcome than a successful-but-rational base implies.	Neutron scales; owned-constellation second act begins. WHY 22% A chain: Neutron reusability works AND captures share AND Space Systems scales AND an owned constellation begins generating recurring revenue. Each plausible at 50-70%; jointly ~22%. RKLБ has the balance sheet, backlog, and vertical integration — but it is conjunctive. WHAT HAPPENS The bull: Neutron's reusability lands RKLБ real medium-lift share, Space Systems becomes a top-tier prime, AND RKLБ begins operating its own constellation — recurring, higher-margin data/services revenue layered on launch+systems. Revenue compounds toward ~\$16.5B by FY35 at a ~31% blended operating margin. Minimal dilution; strong self-funding. DCF ~\$82/share — still ~42% below spot, but the owned-constellation second act is now visible in the numbers, not just the story. The launch business is the wedge; the constellation is the business.	Starlink-style constellation; smaller-SpaceX by 2035. WHY 8% Tail of tails: every bull condition hits AND RKLБ originates an owned constellation at platform scale AND earns a platform terminal as the network-economies Power lands. Individually 50-65%; jointly ~8%. The Starlink analog is the anchor; the \$6d constellation lead/lag metrics are the falsifiers. WHAT HAPPENS The tail (the second act): RKLБ originates a Starlink-style owned constellation on top of a vertically-integrated launch+systems prime — recurring data/services revenue at platform margins. Revenue reaches ~\$36.5B by FY35 at a ~38% blended margin, with a platform terminal reflecting the originated network-economies Power (\$6d). DCF ~\$367/share — equity ~\$213B, a credible smaller-SpaceX (~5.8x sales, vs SpaceX's ~23x private mark, most of which is Starlink). The genuine exponential tail, and it finally clears spot (+159%): the value is the constellation, not the rockets.

Rocket Lab business snapshot

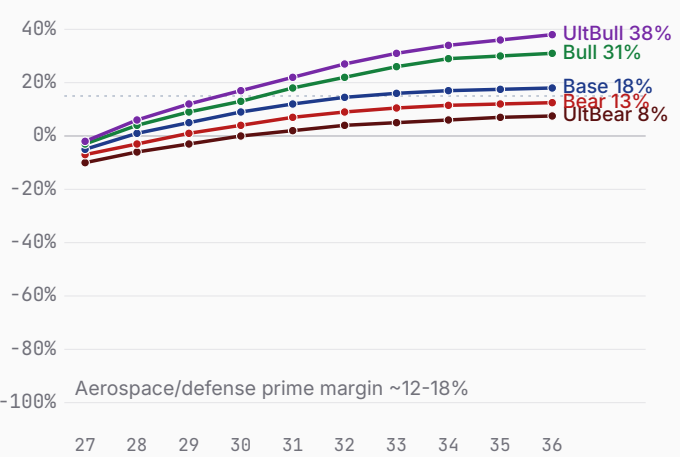
Bear \$3.70 Base \$11.65 Bull \$82.02

FY23-FY25 history + FY26-FY35 scenario projections · fiscal years end Dec 31 · 10-K FY25, Q4'25 release

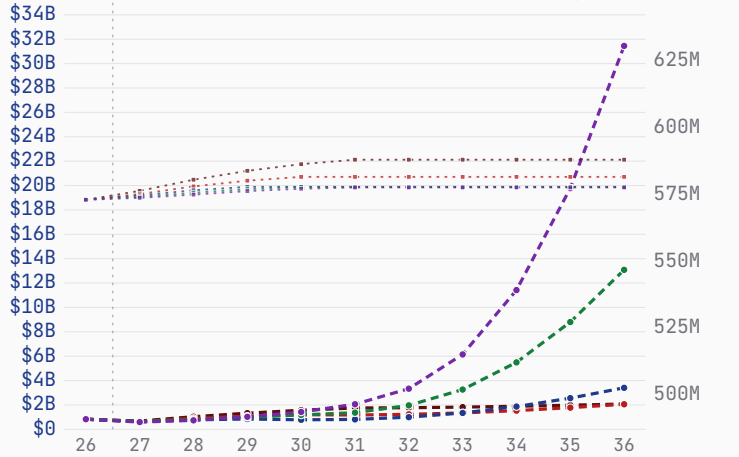
Revenue — history + scenarios to FY36 (log)



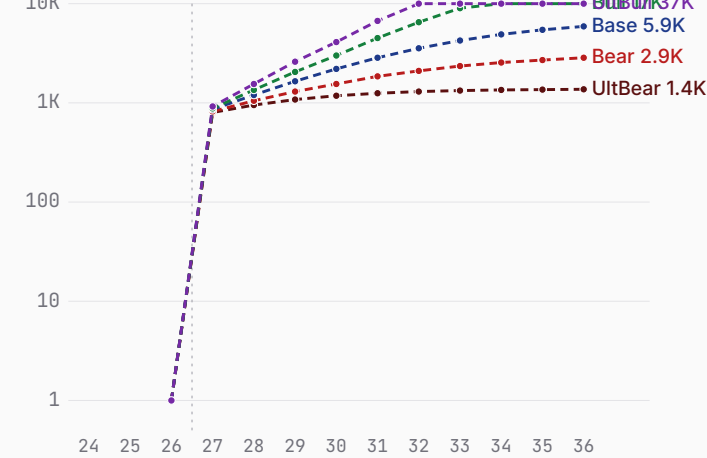
Path to profitability — op margin (FY27→FY36)



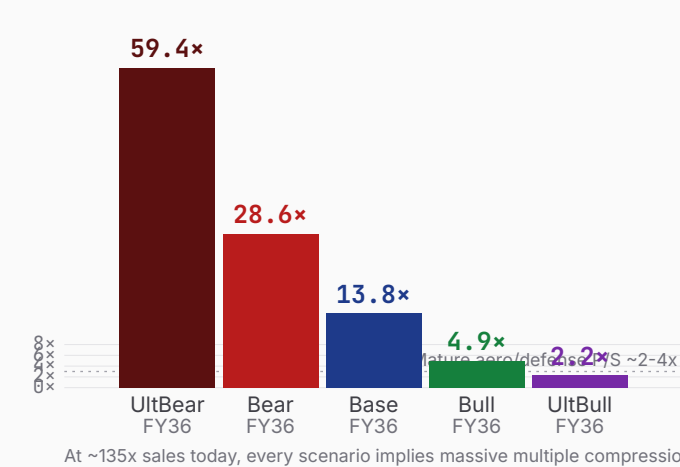
Cash + dilution (FY26 → FY36)



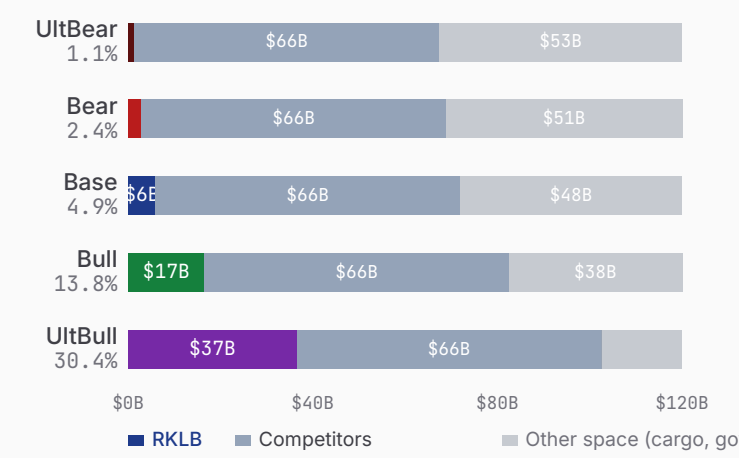
Revenue ramp (log)



\$81.40B mkt cap as P/S on FY36 rev



\$120B FY35 launch + space-systems TAM — RKL share



Sources: RKL FY2025 results (Feb 2026), Q4'25 8-K, company guidance. FCF is NOL-shielded pre-tax over the explicit window (documented simplification). TAM: launch + space-systems serviceable slice, ~\$120B by FY35.

Rocket Lab 7 Powers · the moat, scored

Bear \$3.70 Base \$11.65 Bull \$82.02

Arena. Space launch + space systems — Rocket Lab (Electron small-lift, Neutron medium-lift reusable coming, vertically-integrated Space Systems) vs SpaceX (dominant), Firefly, Stoke Space, ULA/Blue Origin; the race to become a vertically-integrated space prime.

POWER ORIENTATION
origination window: open

POWER ORIENTATION · 7 POWERS

Scale Economies	Electron cadence builds a learning curve; Neutron reusability could reset cost, but it is pre-launch.
Network Economies	Owned-constellation/data flywheel — the ultra-bull second act (\$6c.11.1); being originated, not yet established.
Counter-Positioning	Vertically integrated launch + components + future constellation — few new-space peers can replicate the full stack.
Switching Costs	Space Systems components get designed-in (some lock-in); launch is largely contestable.
Branding	The credible #2 launch brand after SpaceX; strong gov/defense trust.
Cornered Resource · thesis leans here	Launch sites/licenses, an acquired vertically-integrated component portfolio, Beck-led engineering talent, Neutron IP.
Process Power	Electron reliability + manufacturing learning curve; a genuine, if narrow, process edge.

COMPETITIVE READ

Rocket Lab is originating a genuine vertically-integrated space-prime position — the credible #2 to SpaceX — with real Space Systems scale, a \$1.85B backlog, and Neutron optionality. But it lags SpaceX by an order of magnitude on launch cadence/reusability, and Neutron is unproven; the window is open as the #2, but SpaceX's cost-and-scale gravity well is the falsifier the whole bull case must clear.

THE RACE · NAMED RIVALS

SpaceX	\$ tens of B; Starlink cash flow
~45% terminal share · Falcon 9 cadence + Starship + Starlink — the gravity well of the industry.	
ULA / Blue Origin	Boeing/LM + Bezos
~12% terminal share · Vulcan / New Glenn; legacy contracts + Bezos capital.	
Firefly Aerospace	public; funded
~8% terminal share · Alpha + Elytra + MLV; lunar lander; recently public.	
Stoke Space	~\$ hundreds M
~5% terminal share · Fully reusable from day one; early but ambitious.	

LEAD / LAG · FALSIFIERS

Medium-lift reusable launch	Neutron pre-launch vs Falcon 9 + Starship	LAGGING
Launch cadence (per yr)	~15-20 (Electron) vs >100 (SpaceX)	LAGGING
Space Systems revenue scale	leading new-space vs fragmented	LEADING

Rocket Lab show your work

Bear \$3.70 Base \$11.65 Bull \$82.02 · Weighted \$52.04 (-63.4%)

ULTRA BEAR

\$1.80

BEAR

\$3.70

BASE

\$11.65

BULL

\$82.02

ULTRA BULL

\$360.14

ASSUMPTIONS

TAM Year-10 (\$B)	120
Mature share (%)	1.1
Mature op margin (%)	8
Sales-to-capital	1.7
Terminal WACC (%)	11.0
Terminal growth (%)	2.5
P(failure) (%)	25
Scenario probability (%)	10

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.80	-10%	-0.20	-0.17
FY28	0.95	-6%	-0.14	-0.11
FY29	1.08	-3%	-0.11	-0.07
FY30	1.18	+0%	-0.06	-0.04
FY31	1.25	+2%	-0.02	-0.01
FY32	1.30	+4%	+0.02	+0.01
FY33	1.33	+5%	+0.05	+0.02
FY34	1.35	+6%	+0.07	+0.03
FY35	1.36	+7%	+0.09	+0.03
FY36	1.37	+8%	+0.10	+0.03

Σ PV explicit FCF -0.28

+ PV terminal (g 2.5%) 0.38

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$0.10B
+ Cash & investments	\$0.83B
= Equity value	\$0.93B
Raised over period	\$1.80B
Dilution by FY36	+3%
FY36 shares (M)	588

DCF per share	\$1.58
× (1-P_fail) [75%]	\$1.19
+ P_fail × distress [25% × \$2.50]	\$0.63

= Expected per share \$1.80

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$3.03	\$5.11	\$8.61	\$14.51
0.02×	0.04×	0.06×	0.10×

ASSUMPTIONS

TAM Year-10 (\$B)	120
Mature share (%)	2.4
Mature op margin (%)	13
Sales-to-capital	1.9
Terminal WACC (%)	10.2
Terminal growth (%)	3.0
P(failure) (%)	12
Scenario probability (%)	25

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.82	-7%	-0.17	-0.15
FY28	1.05	-3%	-0.15	-0.12
FY29	1.30	+1%	-0.12	-0.09
FY30	1.55	+4%	-0.07	-0.04
FY31	1.85	+7%	-0.03	-0.02
FY32	2.10	+9%	+0.06	+0.03
FY33	2.35	+11%	+0.12	+0.05
FY34	2.55	+12%	+0.19	+0.08
FY35	2.70	+12%	+0.24	+0.09
FY36	2.85	+13%	+0.28	+0.10

Σ PV explicit FCF -0.07

+ PV terminal (g 3.0%) 1.37

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$1.30B
+ Cash & investments	\$0.83B
= Equity value	\$2.13B
Raised over period	\$1.20B
Dilution by FY36	+2%
FY36 shares (M)	582

DCF per share	\$3.66
× (1-P_fail) [88%]	\$3.22
+ P_fail × distress [12% × \$4.00]	\$0.48

= Expected per share \$3.70

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$6.01	\$9.77	\$15.88	\$25.81
0.04×	0.07×	0.11×	0.18×

ASSUMPTIONS

TAM Year-10 (\$B)	120
Mature share (%)	4.9
Mature op margin (%)	18
Sales-to-capital	2.1
Terminal WACC (%)	9.5
Terminal growth (%)	3.5
P(failure) (%)	5
Scenario probability (%)	35

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.86	-5%	-0.17	-0.15
FY28	1.20	+1%	-0.15	-0.12
FY29	1.65	+5%	-0.13	-0.10
FY30	2.20	+9%	-0.06	-0.04
FY31	2.85	+12%	+0.03	+0.02
FY32	3.55	+14%	+0.18	+0.10
FY33	4.25	+16%	+0.35	+0.17
FY34	4.90	+17%	+0.52	+0.23
FY35	5.45	+18%	+0.69	+0.28
FY36	5.90	+18%	+0.85	+0.31

Σ PV explicit FCF +0.70

+ PV terminal (g 3.5%) 5.38

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$6.08B
+ Cash & investments	\$0.83B
= Equity value	\$6.91B
Raised over period	\$0.75B
Dilution by FY36	+1%
FY36 shares (M)	578

DCF per share	\$11.96
× (1-P_fail) [95%]	\$11.36
+ P_fail × distress [5% × \$6.00]	\$0.30

= Expected per share \$11.65

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$18.34	\$28.87	\$45.45	\$71.55
0.13×	0.20×	0.32×	0.50×

ASSUMPTIONS

TAM Year-10 (\$B)	120
Mature share (%)	13.8
Mature op margin (%)	31
Sales-to-capital	2.4
Terminal WACC (%)	9.0
Terminal growth (%)	5.0
P(failure) (%)	3
Scenario probability (%)	22

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.88	-3%	-0.14	-0.13
FY28	1.35	+4%	-0.14	-0.12
FY29	2.05	+9%	-0.11	-0.08
FY30	3.00	+13%	-0.01	-0.00
FY31	4.50	+18%	+0.18	+0.11
FY32	6.50	+22%	+0.60	+0.33
FY33	9.00	+26%	+1.30	+0.65
FY34	12.00	+29%	+2.23	+1.02
FY35	14.50	+30%	+3.31	+1.39
FY36	16.50	+31%	+4.28	+1.65

Σ PV explicit FCF +4.82

+ PV terminal (g 5.0%) 43.23

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$48.05B
+ Cash & investments	\$0.83B
= Equity value	\$48.88B
Raised over period	\$1.00B
Dilution by FY36	+0%
FY36 shares (M)	580

DCF per share	\$84.28
× (1-P_fail) [97%]	\$81.75
+ P_fail × distress [3% × \$9.00]	\$0.27

= Expected per share \$82.02

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$126.20	\$194.17	\$298.76	\$459.67
0.89×	1.37×	2.10×	3.24×

ASSUMPTIONS

TAM Year-10 (\$B)	120
Mature share (%)	30.4
Mature op margin (%)	38
Sales-to-capital	2.5
Terminal WACC (%)	8.5
Terminal growth (%)	6.0
P(failure) (%)	2
Scenario probability (%)	8

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.92	-2%	-0.15	-0.13
FY28	1.55	+6%	-0.16	-0.13
FY29	2.60	+12%	-0.11	-0.08
FY30	4.10	+17%	+0.10	+0.07
FY31	6.70	+22%	+0.43	+0.27
FY32	10.80	+27%	+1.28	+0.72
FY33	16.80	+31%	+2.81	+1.45
FY34	24.00	+34%	+5.28	+2.51
FY35	31.00	+36%	+8.36	+3.65
FY36	36.50	+38%	+11.67	+4.70

Σ PV explicit FCF +13.02

+ PV terminal (g 6.0%) 199.15

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$212.17B
+ Cash & investments	\$0.83B
= Equity value	\$213.00B
Raised over period	\$1.35B
Dilution by FY36	+0%
FY36 shares (M)	580

DCF per share	\$367.24
× (1-P_fail) [98%]	\$359.90
+ P_fail × distress [2% × \$12.00]	\$0.24

= Expected per share \$360.14

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$541.53	\$814.27	\$1,224.38	\$1,841.05
3.81×	5.73×	8.62×	12.97×



Rocket Lab People · Opportunity · Context · Deal

Bear \$3.70 Base \$11.65 Bull \$82.02

Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Sir Peter Beck · founder-led13 yrs in seat

10%insider ownershipHIGHkey-person risk

Capital allocation (realized)
Vertical-integration roll-up (Space Systems): six closed acquisitions — Sinclair, ASI, Planetary Systems, SolAero (\$80M), Geost (\$275M), Mynaric (\$155M) — funded partly by a \$355M 4.25% 2029 convertible (2024). No buyback; share count grows with organic + stock-funded-M&A dilution.

Incentive alignment
Equity-heavy comp (2024 ~\$20M, ~97% equity). March 2026: Beck cut salary to \$1, waived bonus, and forfeited all unvested RSUs — alignment now rests on his ~10% stake. RSUs time-based (no confirmed TSR/PSU program).

GOVERNANCE FLAGS

- SPAC origin (Vector Acquisition, 2021)
- CEO + Chairman combined (lead independent director)
- Series A founder control-continuity preferred (economically pass-through)
- majority-independent board (single insider)

PEOPLE READ

Founder-owner-operator (~10%, salary cut to \$1 in 2026) with a majority-independent board — strongly aligned; the offsets are acute key-person concentration (Beck is CEO / Chair / de-facto CTO / public face) and steady dilution from stock-funded M&A.

THE FOUR LEGS

Peopleobservable composite · 1–5

Opportunity\$6c scenarios · Page-3 business snapshot

Context\$6d 7 Powers (Power Origination)

DealOpen-market purchase = the deal: entry price vs. the \$6c scenario distribution (the –63% finding) + \$12 sizing.



Rocket Lab supporting analysis · disclaimers · glossary

Bear \$3.70 Base \$11.65 Bull \$82.02

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1 **Best-capitalized new-space name.**
\$0.83B cash + \$1.85B backlog + ATM access; far more de-risked than pre-revenue peers.

4 **Reusability could reset the cost curve.**
If Neutron lands reuse economics, the bull launch-share path opens — the single biggest swing factor.
- 2 **Space Systems is the hidden engine.**
The majority of revenue is components/satellites, not launch — diversified, growing, and sticky (designed-in).

5 **Backlog visibility.**
\$1.85B backlog (+73% YoY) gives unusual revenue visibility for a 'young' name.
- 3 **Neutron is optionality, not the whole thesis.**
Even modeling Neutron conservatively, Space Systems alone compounds to a real business.

FALSIFICATION TRIGGERS

Bull validation Neutron reaches orbit on schedule (Q4'26) · reuse demonstrated by FY27 · Space Systems > \$1B annual run-rate	Bear validation Neutron slips beyond FY27 · gross margin stalls < 35% · Space Systems growth decelerates < 20%	Reframe needed If reusable medium-lift cadence approaches Falcon 9 economics, the launch-share ceiling lifts materially — revisit the bull tam_share
---	--	--

DISCLAIMERS · PLEASE READ BEFORE USING THIS DOCUMENT

Not investment advice. This document is produced for entertainment and educational purposes only by an individual amateur investor. "Alameda Research 2: Electric Boogaloo" is the unincorporated personal concept of one person and is not a registered investment management entity. Nothing herein constitutes investment advice, a recommendation, or a solicitation to buy or sell any security.	Not a professional. The author is not a registered investment advisor, broker-dealer, CFA charterholder, certified financial planner, or licensed financial professional of any kind. The author has no formal training in equity research or capital markets. No fiduciary relationship is created by reading this document.	AI-assisted analysis. Substantial portions of this analysis were produced with the assistance of large language models. LLMs are known to fabricate facts, miscalculate numbers, hallucinate sources, and present incorrect information with high apparent confidence. Every figure, claim, and projection in this document should be independently verified before acting on it.
Forward-looking statements. Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.	Author may hold positions. The author may own, intend to acquire, or hold short exposure to \$RKLB or related securities at any time. Positions may change without notice and without an update to this document. Do not assume the author's positions match the tone of the analysis.	Do your own research. Do not make investment decisions on the basis of this document. Consult a licensed financial professional and read the company's official SEC filings (10-K, 10-Q, 8-K, proxy) before forming any investment view. If you wouldn't trust your retirement to a chatbot, don't trust this either.

GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Electron — RKLB's small-lift (~300kg) orbital rocket; the established #2 to SpaceX by launch count.	Neutron — RKLB's reusable medium-lift (~13t) rocket targeting a Q4 2026 first launch; the core bull catalyst.	Space Systems — RKLB's non-launch segment: satellites, star trackers, solar, separation systems, components — the majority of revenue.
Backlog — Contracted future revenue not yet recognized; RKLB's was \$1.85B (+73% YoY) at FY25.	Sales-to-capital (s2c) — Damodaran reinvestment efficiency: incremental revenue per dollar of reinvested capital. Higher = more capital-efficient growth.	